

YOLO v2:

- Improved version of YOLOv1 + Prior boxes (from other approaches) + novel training method
- multi-scale training.
- Jointly train on classification & det. datasets. (Weakly supervise BBox predictions using classification dataset for which we have no BBox labels)
- Motive: cls. datasets ↑ variety (categories) and count ↑ (Millions compared to 10k in Detection)
- make use of cls. data to ↑ Detection results.

Better:

- Drawback of YOLO v1: Poor localisation (Poor recall)
- failed to detect small objects

Targets: ↑ recall, localisation acc
Simplify network to make learning easier.
representation

DBN: ↑ convergence, faster, a regularization method.
2% mAP ↑ ~ remove dropout

Imagenet, then finetuning 448 img for detection

Now: After training 224×224 for cls, finetuning model for 448×448 imgs in Imagenet cls. itself for vlocp. That too $\rightarrow$ for detection tasks.

③ Conv. with anchor boxes:

$\rightarrow$ YOLO used to predict BBox directly.

$\rightarrow$ But Faster R-CNN ^{uses} anchor boxes (predefined, computed)

model predicts Δ coordinates w.r.t. anchor boxes.

$\rightarrow$ Predicting offsets instead of coordinates easier

$\rightarrow 448 \rightarrow 416$ $416 // 32 = 13 \times 13$ grid. (odd grid number coz we want grid cell @ centre)

$\rightarrow$ Class, prob score now coupled with anchor boxes, instead of grid cell

$\rightarrow$ YOLOv1 $\Rightarrow$ 98 Boxes/image, YOLOv2 $\Rightarrow$ > 1K boxes

$\rightarrow$ mAP from 69.5 $\rightarrow$ 69.2 but recall image $(81\% \rightarrow 85\%)$

④ Dimension clusters:

$\rightarrow$ Instead of using hardcoded priors for anchor boxes, estimate using clustering in dataset for

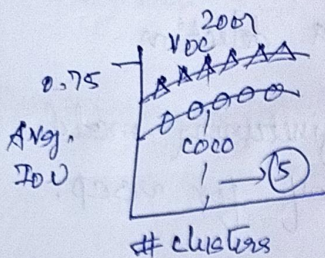
$\rightarrow$ K-means clustering on tgt Box size and aspect ratio

$\rightarrow$ distance metric for K-means!

$$d(\text{box}, \text{centroid}) = 1 - \text{IOU}(\text{box}, \text{centroid})$$

$\swarrow$ chosen
b/w anchor boxes
and all tgt. Boxes

As IOU $\uparrow$, $d \downarrow$, inverse relationship



Yes, tradeoff b/w complexity & recall

Better than prior anchor boxes (short, wide)

Box Generation	#	Avg IOU	with 5 anchor boxes itself - mat (cho)
cluster SE	5	58.7	baseline
cluster IOU	5	61.0	
Anchor Boxes	9	60.9	$\Rightarrow$ Faster FCNN baseline
cluster IOU	9	67.2	

⑤ Direct location prediction:

$\rightarrow$ Anchor box usage $\rightarrow$ divergence, instability during early training.

General formulation:

$$\begin{aligned} x &= (t_x * w_x) - x_a \\ y &= (t_y * w_y) - y_a \end{aligned}$$

$\rightarrow$ anchor box values

t_x, t_y = unconstrained
Hence any anchor box can theoretically predict obj from any grid cell

offsets model's trying to predict.

So: Coordinates relative to grid cell.

$$\begin{aligned} b_x &= \sigma(t_x) + c_x \\ b_y &= \sigma(t_y) + c_y \\ b_w &= p_{w_{\text{net}}} \\ b_h &= p_{h_{\text{net}}} \\ p_{\text{object}} + \text{IOU}(b, \text{object}) &= \sigma(t_o) \end{aligned}$$

c_x, c_y = cell top left corner. i.e. i.e. using sigmoid (logit) reg. $\rightarrow$ max value within grid cell

∴ for each anchor box in grid cell, we produce 5 values t_x, t_y, t_w, t_h and t_o

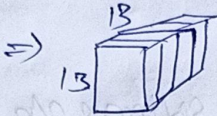
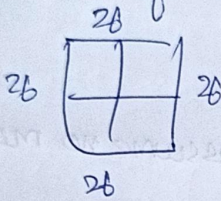
Dimension clusters + Direct location pred $\uparrow$ 5% over YOLO V1

(6)

Fine grained features:

→ Small objects require combine feat. maps from prev. layers. (as in Faster RCNN, SSD etc).

→ Instead of residual connection - divide and concatenate channels



$$\therefore 26 \times 26 \times 512 \Rightarrow 13 \times 13 \times 2048$$

1% $\uparrow$ over YOLO V1

This is ~~at~~ stacked with small res. features.

(7) Multi-scale training:

Every 10 batches, training size sampled from {320, 352, 384, ..., 608} set. Robust to multiple scale.

Faster:

→ Most used VGG16 as backbone (30.69 bn FLOPs)

→ Custom ResNet backbone for YOLO 224x224 (88% top-5 accuracy)

→ Darknet-19 GoogLeNet inspired (3x3, 1x1, Global Avg. Pooling)

→ Imagenet Pretrain: 224x224 μ 160 ep / 0.1 poly LR schedule
SGD, 0.9 momentum, 0.0005 weight decay, 4

Augmentation: random crops, rotations, HSV changes

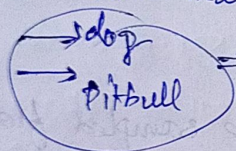
changing to ~~416~~ 416 x 416 res. on Imagenet $\rightarrow$ top 5% accuracy
 $\boxed{= 93.5\%}$

Detection Training:

VOC: final layer: (20 classes) $(5 + C)$ * number anchor boxes
 $(25) * 5$
 $\boxed{= 125 \text{ values per grid cell}}$

Joint Detection & classification training

- $\rightarrow$ most cls. approaches = softmax o/p (assume no multi-label mutually exclusive classes)
- $\rightarrow$ Problem when combining datasets.
- $\rightarrow$ If we assume multilabel, then structure of data is lost



$\Rightarrow$ considered 2 separate labels but in reality, there's a structure.

Soln: Hierarchical classification using WordTree

YOLOv3: Incremental improvements:

- $\rightarrow$ Tech report format.
- $\rightarrow$ Integrate ideas from other models + New Backbone
- $\rightarrow$ Bounding Box Prediction:

$\rightarrow$ same as YOLOv2, using prior anchor boxes (computed using Kmeans clustering) and predicting offsets.

$$\begin{aligned} b_x &= \sigma(t_x) + c_x \\ b_y &= \sigma(t_y) + c_y \\ b_w &= p_w e^{t_w} \\ b_h &= p_h e^{t_h} \end{aligned}$$

c_x, c_y = cell top left corner

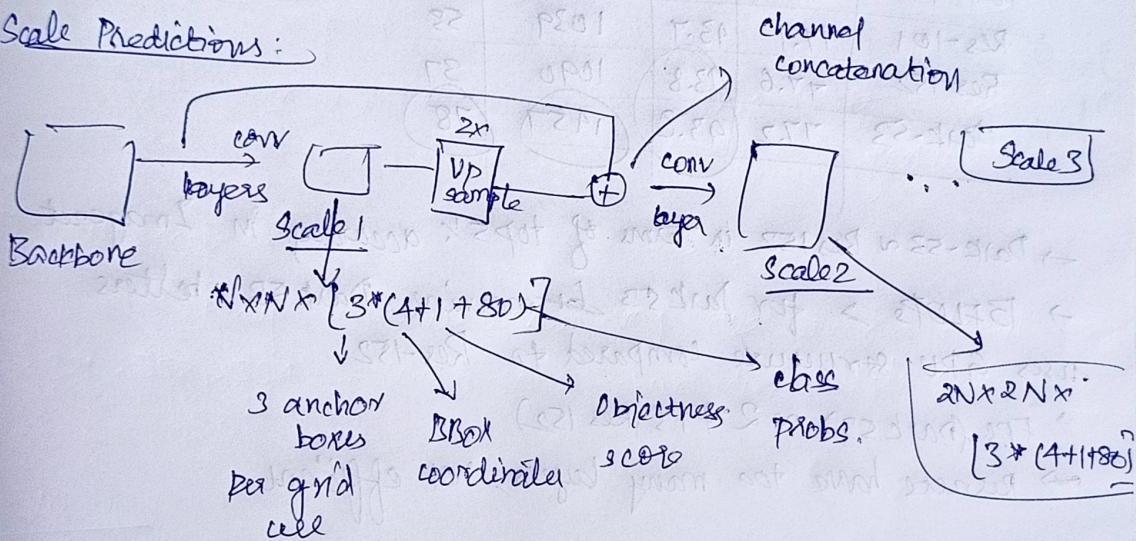
p_w, p_h = anchor box height, width

With SSE based loss fn, $\left[\frac{\partial L}{\partial x} (x^* - \hat{x}) \right]$

Loss gradient $\propto \left(\begin{matrix} \text{original} \\ \text{BBox} \end{matrix} - \begin{matrix} \text{estimated} \\ \text{BBox} \end{matrix} \right)$

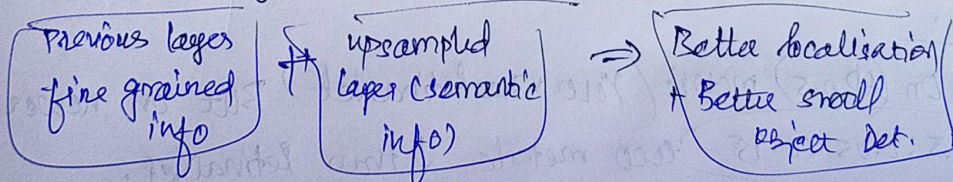
- We now predict 3 anchor boxes per grid cell
- The anchor box with max IOU with gt BBox is responsible others are ignored even if IOU ~~threshold~~ → Threshold
- If gt box has no prior (anchor box with IOU > threshold), then that gt box doesn't contribute to loss
- IOU threshold = 0.5

Scale Predictions:



→ For small object localisation, multi-scale predictions as in FPN.

→ But instead of residual connections, channel-wise concatenation



Scale 1 ⇒ Predictions + Scale 2 Predictions + Scale 3 Predictions = Total Predictions

Now 9 prior anchor boxes, (3 for each scale).

(10x13) (16x30), (33x23), (30x61), (62,45), (59x119),
(116x90), (156x198), (373x326)

→ classification - 80 classes separate ~~softmax~~ sigmoid + BCE loss fn
→ helpful when extending to multi-label datasets.

Backbone:

→ Darknet-53 with Res. connections + YOLOv2 stuff + Dropout - 19

Backbone	Top1	Top5	BFLOP%	FPS
Dark-19	74.7	91.8	1246	171
Res-101	77.1	93.7	1039	58
Res-152	77.6	93.8	1090	37
Dark-53	77.2	93.8	1457	78

→ Dark-53 ~ Res-152 in terms of top5% accuracy in ImageNet

→ BFLOP > for Dark-53 ~~for~~ meaning Dark-53 better
uses GPU structure compared to Res-152

→ FPS (Dark-53) ~ 2 * FPS (Res-152)

→ Resnets have too many layers, not efficient

Training:

→ No -ve hard mining, ~~not~~

→ Multi-scale training - data augmentation, BN

Results:

→ On (AP_{0.5}) metric (YOLO v3 = RetinaNet) But on newer
mAP_{0.5:0.05:0.95} coco metric ↓ than RetinaNet.

→ At high IoU (IDs) ↓ AP ⇒ struggles to get perfect
bounding boxes

COCO results.

model	Backbone	AP	AP ₅₀	AP ₇₅	AP ₀	AP _m	AP _L
RetinaNet	ResNet 101 FPN	40.8	61.0	44.1	24.1		
YOLO v3	Dark 53	33.0	57.9	34.4	18.3	44.2	51.2
						35.4	41.9

Things that didn't work:

→ Direct anchor box (x, y) predictions

→ Anchor box offset using linear activations (no σ) and scaled ~~weight~~ of box width, height

→ Focal loss: YOLO v3 may be already robust to problem focal loss solves, because it has separate objectness prediction and conditional class probabilities